

Poisson Integral Formula: Disk and Upper Half-Plane

The Dirichlet Problem (General Domain)

Let $D \subset \mathbb{R}^2$ be a domain with boundary ∂D .

Given boundary data $f : \partial D \rightarrow \mathbb{R}$, find $u : \overline{D} \rightarrow \mathbb{R}$ such that

$$\Delta u = 0 \text{ in } D \text{ and } u = f \text{ on } \partial D.$$

Questions:

- ▶ Existence: does a solution exist?
- ▶ Uniqueness: is the solution unique?
- ▶ Representation: can u be written explicitly in terms of f ?

Goal of this lecture: Construct explicit solutions in special domains (disk and upper half-plane) via Poisson kernels.

Harmonic Functions on the Disk

Let u be harmonic in the unit disk

$$D = \{z : |z| < 1\}$$

and continuous on $\overline{D}$.

Goal: recover u from its boundary values.

Complex Integral Formula (Cauchy / Poisson Lead-in)

Recall the Cauchy integral formula for analytic f on D :

$$f(z) = \frac{1}{2\pi i} \int_{|\zeta|=1} \frac{f(\zeta)}{\zeta - z} d\zeta, \quad |z| < 1.$$

Write $z = re^{i\theta}$ and $\zeta = e^{i\phi}$ so that $d\zeta = ie^{i\phi} d\phi$:

$$f(z) = \frac{1}{2\pi} \int_0^{2\pi} \frac{f(e^{i\phi})}{e^{i\phi} - z} e^{i\phi} d\phi.$$

Now take real parts. If $u = \Re f$, then

$$u(z) = \frac{1}{2\pi} \int_0^{2\pi} \Re \left(\frac{e^{i\phi} + z}{e^{i\phi} - z} \right) u(e^{i\phi}) d\phi.$$

Key point: The real part of a simple pole at $\zeta = z$ reconstructs u from the boundary values on a circle.

Poisson Kernel (Disk)

The Poisson kernel is

$$P_r(\theta) = \frac{1 - r^2}{1 - 2r \cos \theta + r^2}.$$

Poisson integral formula:

$$u(re^{i\theta}) = \frac{1}{2\pi} \int_0^{2\pi} \frac{1 - r^2}{1 - 2r \cos(\theta - \phi) + r^2} u(e^{i\phi}) d\phi.$$

Complex Form of the Kernel

Key identity:

$$P_r(\theta - \phi) = \Re \left(\frac{e^{i\phi} + z}{e^{i\phi} - z} \right), \quad z = re^{i\theta}.$$

Compute:

$$\frac{e^{i\phi} + z}{e^{i\phi} - z} = \frac{1 + ze^{-i\phi}}{1 - ze^{-i\phi}}.$$

Multiply numerator and denominator by $1 - \bar{z}e^{i\phi}$.

Algebra for Kernel

Let $z = re^{i\theta}$. Then

$$\begin{aligned}|e^{i\phi} - z|^2 &= (e^{i\phi} - z)(e^{-i\phi} - \bar{z}) \\&= 1 - ze^{-i\phi} - \bar{z}e^{i\phi} + |z|^2 \\&= 1 - 2r \cos(\theta - \phi) + r^2.\end{aligned}$$

Also,

$$\Re \left(\frac{e^{i\phi} + z}{e^{i\phi} - z} \right) = \frac{1 - |z|^2}{|e^{i\phi} - z|^2}.$$

Approximate Identity

Why does this work?

Properties of P_r :

- ▶ $P_r(\theta) > 0$
- ▶ $\frac{1}{2\pi} \int_0^{2\pi} P_r(\theta) d\theta = 1$
- ▶ Concentrates at $\theta = 0$ as $r \rightarrow 1$

Map to Upper Half-Plane

Use the Möbius transformation

$$\phi(\zeta) = i \frac{1 + \zeta}{1 - \zeta}.$$

Inverse:

$$\zeta = \frac{z - i}{z + i}.$$

Maps:

- ▶ Unit disk $\rightarrow$ upper half-plane
- ▶ Unit circle $\rightarrow$ real line

Boundary Parametrization

Let $\zeta = e^{i\phi}$. Then

$$\begin{aligned} t &= \phi(e^{i\phi}) \\ &= i \frac{1 + e^{i\phi}}{1 - e^{i\phi}} \\ &= i \frac{e^{-i\phi/2} + e^{i\phi/2}}{e^{-i\phi/2} - e^{i\phi/2}} \\ &= i \frac{2 \cos(\phi/2)}{-2i \sin(\phi/2)} \\ &= \cot\left(\frac{\phi}{2}\right). \end{aligned}$$

Jacobian Computation

Compute $dt/d\phi$ for $t = \cot(\phi/2)$

$$\frac{dt}{d\phi} = -\frac{1}{2} \csc^2(\phi/2)$$

Thus

$$d\phi = \frac{2}{1+t^2} dt.$$

Transforming the Kernel

We relate disk and half-plane kernels.

Key identity:

$$\frac{1 - r^2}{|e^{i\phi} - re^{i\theta}|^2}$$

Under the map becomes

$$\frac{2y}{(x - t)^2 + y^2}.$$

This follows from substituting

$$z = x + iy = \phi(re^{i\theta}).$$

Final Formula (Upper Half-Plane)

Putting everything together:

$$u(x, y) = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{y}{(x - t)^2 + y^2} u(t, 0) dt.$$

This is the Poisson integral formula on $\mathbb{H}$.

Remarks

- ▶ Translation invariant kernel
- ▶ Natural for Fourier analysis
- ▶ Disk gives upper half plane via Mobius conformal map